

Music Listening Behavior & Weather Analysis

Student	Ali Deha Ozan
Course	DSA 210 — Introduction to Data Science
Institution	Sabanci University
Term	Spring 2026
Repository	github.com/deha-ozan/DSA210projectDeha

Abstract

This report presents a comprehensive data science analysis of the relationship between daily weather conditions in Istanbul and personal music listening behavior derived from Apple Music history. Approximately 93,000 listening records spanning October 2023 to March 2026 were merged with historical weather data from Open-Meteo to create a daily aggregated dataset of 900 observations. Five statistical hypothesis tests and four supervised machine learning models were applied to determine whether weather variables (temperature, precipitation, sunshine hours) can predict listening duration, play count, and skip behavior. The analysis finds that temperature and season significantly correlate with listening duration, while precipitation alone does not. Non-linear models outperform linear regression, and temporal factors (day of week, season) play a meaningful role alongside weather in shaping listening patterns.

1. Introduction

Music listening is an inherently contextual behavior — the songs people choose and the time they spend listening are shaped by mood, schedule, and environment. Weather is one such environmental factor that may plausibly influence listening habits: a rainy afternoon might invite longer, more introspective listening sessions, while a sunny day could lead to shorter, more upbeat engagement.

This project investigates these relationships using a uniquely personal dataset: over two years of Apple Music listening history from a single user in Istanbul, merged with daily weather data for the same period. The analysis proceeds through four phases: data collection and cleaning, exploratory data analysis, hypothesis testing, and machine learning.

Research Questions

- Does precipitation increase daily listening duration?
- Does temperature correlate with how long I listen?
- Does listening activity differ significantly across seasons?
- Do weekends show different skip behavior compared to weekdays?
- Does precipitation correlate with the number of tracks played per day?

2. Data Sources & Preparation

2.1 Apple Music Listening History

Personal listening history was exported from the Apple Data & Privacy portal. The raw dataset contains approximately 93,000 records. Records were filtered to October 2023 through March 2026 (~2.5 years) and aggregated to the daily level, computing total listening time (minutes), number of plays, total skips, skip rate, and fraction of tracks completed per day.

2.2 Weather Data

Daily weather data for Istanbul was retrieved from the Open-Meteo API. Variables collected include mean daily temperature (°C), maximum temperature (°C), total precipitation (mm), and sunshine duration (hours). Binary flags `is_rainy` (precipitation > 1mm) and `is_sunny` were derived, and season was assigned by month.

2.3 Merged Dataset

The two sources were merged on date, producing 900 daily observations. Temporal features `day_of_week`, `is_weekend`, and `month` were engineered from the date column.

Feature	Type	Description
<code>total_listen_min</code>	Target (regression)	Total listening time per day (minutes)
<code>n_plays</code>	Target / feature	Number of tracks played per day
<code>skip_rate</code>	Feature	Proportion of tracks skipped per day
<code>temp_mean</code>	Weather	Mean daily temperature (°C)
<code>temp_max</code>	Weather	Maximum daily temperature (°C)
<code>precipitation</code>	Weather	Total daily precipitation (mm)
<code>sunshine_hours</code>	Weather	Daily sunshine duration (hours)
<code>day_of_week</code>	Temporal	0=Monday ... 6=Sunday
<code>is_weekend</code>	Temporal	Binary: 1 if Saturday or Sunday
<code>month</code>	Temporal	Calendar month (1–12)
<code>is_rainy</code>	Derived	Binary: 1 if precipitation > 1mm

Table 1: Feature set used in hypothesis testing and machine learning.

3. Exploratory Data Analysis

Listening Duration Distribution

Daily listening duration is right-skewed, with a median around 100 minutes and a long tail on high-listening days — consistent with habit-driven behavior that occasionally spikes on long commutes or inactive days.

Seasonal Patterns

Spring (30.0 plays/day) and Autumn (30.0) are the highest-activity seasons, Summer (19.0) is the lowest, and Winter (23.0) falls between. Colder, less active months are associated with more listening.

Day-of-Week Patterns

Weekends show higher mean listening duration and a higher skip rate than weekdays (medians 0.174 vs 0.152), suggesting more exploratory or casual listening on weekends.

Weather Correlations

A Spearman correlation matrix reveals a weak negative correlation between temperature and listening duration ($\rho=-0.215$). Precipitation shows near-zero correlation with play count ($\rho=0.046$).

4. Hypothesis Testing

Five hypotheses were tested at the 5% significance level ($\alpha=0.05$). Non-parametric tests were chosen throughout given the non-normal distribution of listening metrics.

#	Hypothesis	Test	Statistic	p-value	Result
H1	Rainy days lead to more listening time	Mann-Whitney U	U=89245.0	0.0774	Not significant
H2	Temperature correlates with listening duration	Spearman	rho=-0.215	<0.0001	Significant
H3	Listening activity differs across seasons	Kruskal-Wallis	H=55.975	<0.0001	Significant
H4	Weekends have different skip rates	Mann-Whitney U	U=91942.5	0.0081	Significant
H5	Precipitation correlates with play count	Spearman	rho=0.046	0.1655	Not significant

Table 2: Hypothesis test results. Green = significant, Red = not significant.

Key Interpretations

H1 — Not significant	Despite a higher rainy-day median (110.8 min vs 99.6 min non-rainy), the difference is not statistically significant (p=0.0774). Rain alone does not reliably predict longer listening sessions.
H2 — Significant	A weak but significant negative correlation exists between temperature and listening duration (rho=-0.215, p<0.0001). Warmer days are associated with less listening.
H3 — Significant	Listening activity differs significantly across seasons (p<0.0001). Spring and Autumn have the highest median play counts (30.0/day), Summer the lowest (19.0/day).
H4 — Significant	Weekends show a significantly higher skip rate (0.174 vs 0.152, p=0.0081), suggesting more exploratory listening on weekends.
H5 — Not significant	Precipitation shows near-zero correlation with daily play count (rho=0.046, p=0.1655). How many tracks are played is not driven by rain.

A note on H5 (originally planned as valence analysis): The original project proposal included a sixth hypothesis — that song valence (a Spotify audio feature measuring musical positivity) would be lower on rainy days. This was intended to capture a mood-driven weather response beyond listening quantity. Unfortunately, Spotify deprecated the `/v1/audio-features` endpoint for all developer apps created after November 2024, making programmatic retrieval of valence, energy, danceability, and related features completely infeasible for this project. This is a genuine limitation — not a shortcoming of the analysis, but a casualty of an API policy change outside our control. The mood-weather relationship remains an open and interesting question for future work.

5. Machine Learning

Four supervised machine learning models were applied to predict listening duration from weather and temporal features. Two feature sets were used: weather-only (temp_mean, temp_max, precipitation, sunshine_hours) and full (weather + day_of_week, is_weekend, month). An 80/20 train-test split with random_state=42 was used consistently across all models.

5.1 Models

- **Linear Regression** — baseline linear model.
- **Random Forest** — ensemble of decision trees constrained with max_depth=5 and min_samples_leaf=20 to prevent overfitting on the 900-row dataset.
- **K-Nearest Neighbors (k=25, distance-weighted)** — instance-based learner on scaled features.
- **Decision Tree (max_depth=5)** — classification: predict rainy vs non-rainy from listening behavior.

Model	Features	R ²	MAE (min)	RMSE (min)
Linear Regression	Weather only	0.0499	59.20	73.17
Linear Regression	Weather + day-of-week	0.0303	60.49	73.93
Random Forest	Weather only	0.0825	57.20	71.91
Random Forest	Weather + day-of-week	0.0846	57.57	71.83
KNN (k=25)	Weather + day-of-week	0.0365	58.28	73.69

Table 3: Regression model performance on the test set (180 observations).

5.2 Classification: Decision Tree

A Decision Tree classifier (max_depth=5) was trained to predict whether a day was rainy from listening behavior and temporal features. Accuracy was 67.2% — essentially majority-class baseline (70.6% of days are non-rainy). Listening patterns cannot predict weather, confirming the relationship is one-directional.

5.3 Feature Importance

Random Forest feature importance scores show that weather variables dominate: temp_mean, sunshine_hours, and temp_max are the top three features. day_of_week and month contribute meaningfully but rank below the top weather features. This aligns with hypothesis test results — temperature and season are the significant predictors, not precipitation.

5.4 Residual Analysis

Linear Regression residuals show structured curvature, confirming non-linear relationships exist. Random Forest residuals are more evenly distributed around zero. Both models show high residual variance — consistent with the low R^2 values — indicating a large portion of listening variance is unexplained by weather and temporal features alone.

6. Discussion

The results paint a coherent picture: weather has a real but modest influence on music listening behavior. Temperature and season are the primary weather-related predictors, while precipitation shows no statistically significant relationship with listening duration or play count. This may reflect Istanbul's relatively mild climate — rainy days are not dramatically different from cloudy non-rainy days, and do not force the behavioral changes that might drive longer listening.

The low R^2 values across all models (0.03–0.08) indicate that most variance in daily listening is not explained by weather or day-of-week alone. This is a substantive finding: listening is driven by unobserved personal factors such as academic workload, social engagements, and mood. Weather provides a weak signal in the background.

Random Forest consistently outperforming Linear Regression (R^2 0.0825 vs 0.0499 on weather-only features) confirms that the weather-listening relationship is non-linear — simple correlation understates the effect.

Limitations

- Single individual — findings are not generalizable.
- Spotify audio feature analysis was planned but the /v1/audio-features endpoint was deprecated for new developer apps after November 2024.
- 900 daily observations limits model complexity and statistical power.
- No data on commute, academic calendar, or social events — likely confounders.

7. Conclusions

Weather matters, but modestly	Temperature ($\rho=-0.215$) and season ($p<0.0001$) are significant predictors. Warmer months are associated with less listening.
Rain is not a driver	Neither listening duration (H1, $p=0.0774$) nor play count (H5, $p=0.1655$) is significantly related to precipitation.
Schedule structures listening	Weekends show significantly different skip behavior ($p=0.0081$), independent of weather.
Non-linearity exists	Random Forest outperforms Linear Regression on the same features, and residual plots confirm non-linear patterns in the weather-listening relationship.

Listening is hard to predict

All models achieve R^2 below 0.09. Weather and day-of-week explain only a small fraction of listening variance — the behavior is driven by unobserved personal factors.

8. References & Tools

- Apple Data & Privacy portal — personal listening history export
- Open-Meteo (open-meteo.com) — historical weather API for Istanbul
- Python 3.10 — pandas, numpy, scipy, scikit-learn, matplotlib, seaborn
- Google Colab — notebook execution environment
- GitHub (github.com/deha-ozan/DSA210projectDeha) — project repository
- Claude (Anthropic) — assisted with code drafting, test selection, and report structure. All outputs reviewed and adapted by Ali Deha Ozan.